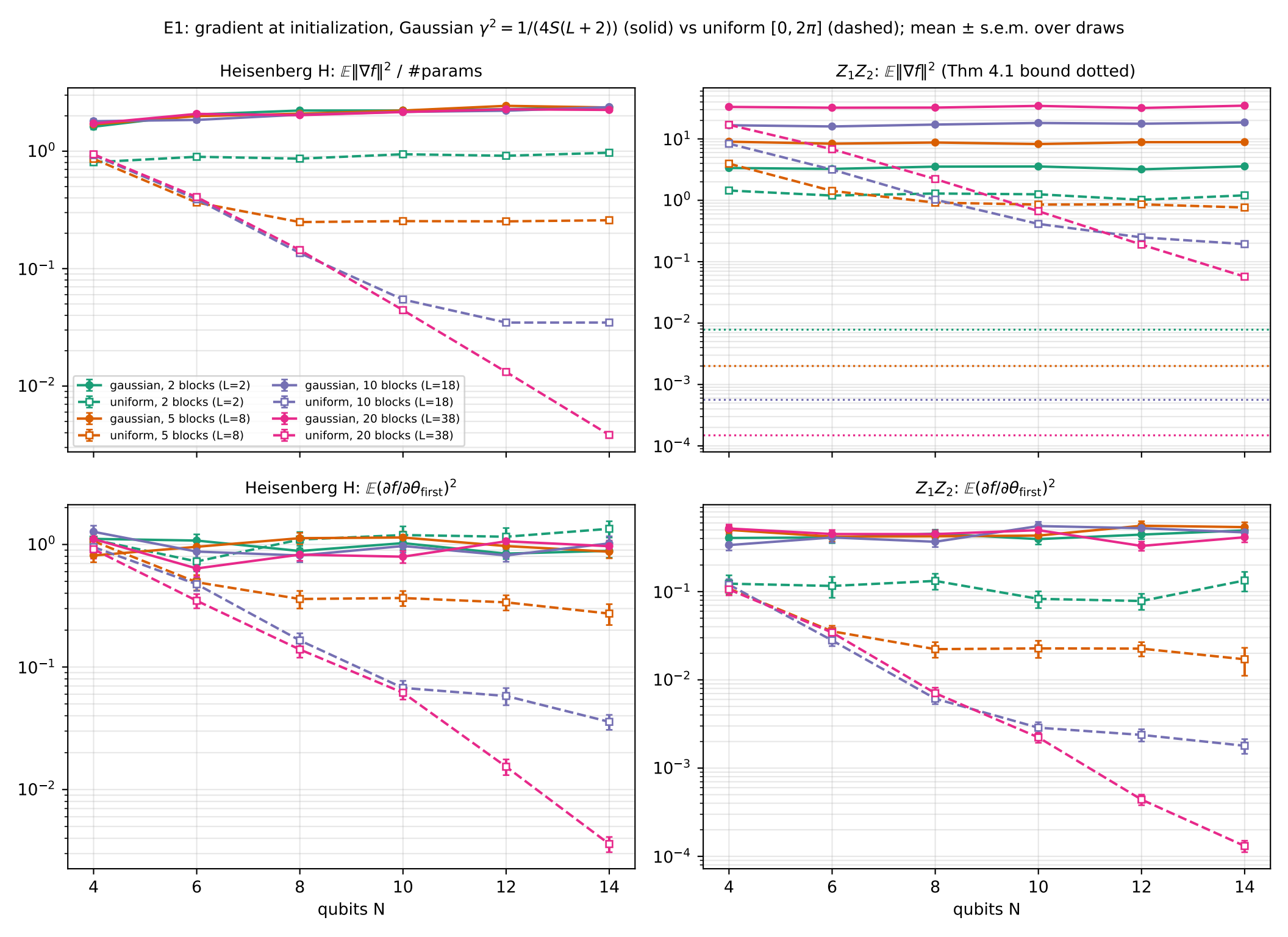
E1: gradient at initialization, Gaussian $\gamma^2 = 1/(4S(L+2))$ (solid) vs uniform $[0, 2\pi]$ (dashed); mean ± s.e.m. over draws
Heisenberg H: $\mathbb{E}\|\nabla f\|^2$ / #params
Z_1Z_2: $\mathbb{E}\|\nabla f\|^2$ (Thm 4.1 bound dotted)
Heisenberg H: $\mathbb{E}(\partial f/\partial\theta_{\text{first}})^2$
Z_1Z_2: $\mathbb{E}(\partial f/\partial\theta_{\text{first}})^2$
gaussian, 2 blocks (L=2)
uniform, 2 blocks (L=2)
gaussian, 5 blocks (L=8)
uniform, 5 blocks (L=8)
gaussian, 10 blocks (L=18)
uniform, 10 blocks (L=18)
gaussian, 20 blocks (L=38)
uniform, 20 blocks (L=38)
qubits N
qubits N